



Unraveling the degradation mechanism of $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ at the high cut-off voltage for lithium ion batteries

Liming Wang^{a,b}, Qingmei Su^{b,d,*}, Bin Han^b, Weihao Shi^{a,b}, Gaohui Du^{b,*}, Yunting Wang^{a,b}, Huayv Li^{a,b}, Lin Gu^{c,*}, Wenqi Zhao^b, Shukai Ding^b, Miao Zhang^b, Yongzhen Yang^{d,e}, Bingshe Xu^{d,e}

^a School of Materials Science & Engineering, Shaanxi University of Science and Technology, Xi'an 710021, Shaanxi, China

^b Materials Institute of Atomic and Molecular Science, Shaanxi University of Science and Technology, Xi'an 710021, Shaanxi, China

^c School of Materials Science & Engineering, Tsinghua University, Beijing 100084, China

^d Shanxi-Zheda Institute of Advanced Materials and Chemical Engineering, Taiyuan 030032, Shanxi, China

^e Key Laboratory of Interface Science and Engineering in Advanced Materials, Taiyuan University of Technology, Taiyuan 030024, Shanxi, China

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ABSTRACT

$\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811) layered oxides have been regarded as promising alternative cathodes for the next generation of high-energy lithium ion batteries (LIBs) due to high discharge capacities and energy densities at high operation voltage. However, the capacity fading under high operation voltage still restricts the practical application. Herein, the capacity degradation mechanism of NCM811 at atomic-scale is studied in detail under various cut-off voltages using aberration-corrected scanning transmission electron microscopy (STEM). It is observed that the crystal structure of NCM811 evolution from a layered structure to a rock-salt phase is directly accompanied by serious intergranular cracks under 4.9 V, which is distinguished from the generally accepted structure evolution of layered, disordered layered, defect rock salt and rock salt phases, also observed under 4.3 and 4.7 V. The electron energy loss spectroscopy analysis also confirms the reduction of Ni and Co from the surface to the bulk, not the previously reported only Li/Ni interlayer mixing. The degradation mechanism of NCM811 at a high cut-off voltage of 4.9 V is attributed to the formation of intergranular cracks induced by defects, the direct formation of the rock salt phase, and the accompanied reduction of Ni^{2+} and Co^{2+} phases from the surface to the bulk.

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1. Introduction

Lithium-ion batteries (LIBs) are widely used in portable electronic products, new energy vehicles and other fields due to their high energy densities, satisfy cycle stability, and safe operation [1–3]. With increasing demands of electrical equipment for the energy and power density of energy storage systems, current LIBs cannot meet the requirements, thus, great efforts have been made on exploring electrode materials with higher reversible capacities and superior cycling stability.

The ternary layered metal (Co, Mn, Ni, Al, etc.) oxide electrode ($\text{LiNi}_{1-x-y}\text{Co}_x\text{Mn}_y\text{O}_2$) with Ni enrichment, with high theoretical specific capacity ($>250 \text{ mA h g}^{-1}$) and higher working voltage ($\sim 3.65 \text{ V}$), has been extensively studied [4,5]. However, there are some inherent problems associated with these materials, such as

the inevitable Li/Ni cation mixing, structural instability, transition-metal dissolution, crack formation and side reactions at the cathode-electrolyte interface [6–10]. All these issues cause the electrochemical performance deterioration of the cathode materials. Previous works reported that there were no significant structural changes of bulk NCM811 cycled between 2.8 and 4.1 V, suggesting the capacity failure cannot only be solely ascribed to structural changes, but electrolyte oxidation was believed to be responsible [11,12]. Great efforts have been focused on improving the electrochemical performance through high Ni content, lattice doping, tuning the compositions and surface modification [13–16]. To further improve the reversible capacity of NCM cathode materials and insert/extract more Li^+ ions, enhancing the cut-off voltage is essential. Unfortunately, a high voltage often means more side reactions between the cathode material and electrolyte interface, phase transition and the generation of intergranular cracks and intragranular cracks [17]. Therefore, it is of great significance to study the structure change of cathode materials under high cut-off voltage and elucidate the degradation mechanism of

* Corresponding authors.

E-mail addresses: suqingmei@sust.edu.cn (Q. Su), dugaohui@sust.edu.cn (G. Du), lingu@mail.tsinghua.edu.cn (L. Gu).

cathode materials during cycles to improve the specific capacity of cathode materials [18]. Kim's group revealed the manifestation of the layered Ni^{2+}O rock-salt phase along with the structural evolution of Ni-rich NCM materials under high-voltage battery operation using XRD, ND, TEM and XAS analyses [19]. Gu et al. tracked the transition metal (TM) migration in O3-type layered NCM811 during a phase transition from the well-order layered (R-3 m) structure into quasi rock-salt (Fm-3 m) structure by in situ aberration corrected scanning transmission electron microscopy (STEM) observation [20]. Yan et al. proposed the electrochemical, thermal, and mechanical coupling effects intimately induced phase inhomogeneity and intragranular cracking, and thus result in an aggravating effect on battery performance using in situ TEM [21]. Although great efforts have been made on the degradation mechanism of NCM cathode, these results were given only in the local region. Few studies have given information on both atomic-level local structure and bulk phase of cathode material in lithium intercalation and degradation of electrochemical performance at high cut-off voltage. In fact, a fundamental understanding of the changes in local structure and bulk phase of ternary layered metal oxide electrode is critical to reveal the origin of capacity fading [22,23]. Furthermore, it is significant to unravel the relationship between the electrochemical performance and the local structure and bulk phase for improving the electrochemical performance of NCM cathode materials by optimizing the compositions and atomic structure.

In this study, with the help of advanced electron microscopic techniques, the degradation mechanism of $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811) cathode is discussed through analysis of the crystal structure changes using STEM-HAADF images, and electronic structural changes on the NCM811 from the surface to the bulk after cycled for 50 times through electron energy loss spectroscopy (EELS) under different cut-off voltage operations.

2. Results and discussion

2.1. Crystallographic structure and microstructure of NCM811

The crystallographic structure of the obtained NCM811 was characterized by XRD, as given in Fig. S1 in Supporting Information, all the diffraction peaks of the XRD could be assigned as the layered hexagonal structure of LiNiO_2 phase (PDF#09–0063), indicating the well-defined rhombohedral $\alpha\text{-NaFeO}_2$ -type layered structure (space group R-3 m) with no trace of impurities. SEM measurements were used to determine the general morphology of the pristine NCM811, showing the obvious secondary spherical morphology of 8–20 μm assembled by densely primary particles (Fig. S2 in Supporting Information). We also used FIB to section NCM811 microspheres for cross-sectional SEM and TEM observations, as displayed in Fig. S3 in Supporting Information. It demonstrates that the secondary microspheres of the NCM811 are densely compacted with few voids inside the microspheres. The TEM image also agrees well with SEM observation, displaying the pristine NCM811 particles are submicronic with diameters of several hundred nanometers. The Ni, Co and Mn elements are homogeneously distributed through the entire particle, examined by the STEM EELS-mappings (Fig. S4 in Supporting Information). Note that the contrast of HAADF-STEM image is proportional to atomic number ($Z^{1.7}$), the heavy atoms can be detected directly through HAADF detector, as shown in Fig. 1(a, d and g). The atomic scale HAADF-STEM images of pristine NCM811 suggest atomic-scale crystal structures along the [1-10], [010] and [-441] zone axes. The heavy TM atomic columns can be confirmed from the structure models of [1-10], [010] and [-441] zone axes simulated in Fig. 1(c, f and i). In addition, the selected area electron diffraction (SAED)

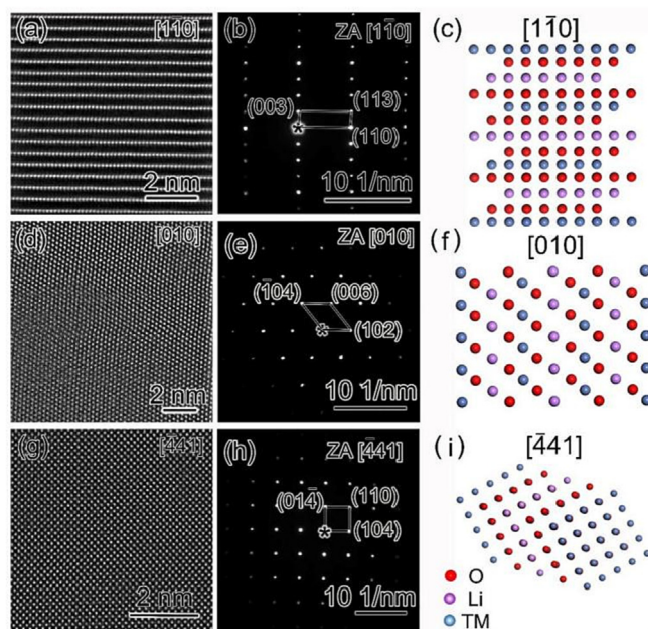


Fig. 1. Atomic-resolution HAADF-STEM images of pristine NCM811 cathode along different orientations. (a, d and g) HAADF-STEM images along [1-10], [010] and [-441] zone axes, respectively. (b, e and h) The corresponding SAED patterns of Figs. (a, d and g). (c, f and i) Structure models of NCM811 along [1-10], [010] and [-441] zone axes.

patterns demonstrated in Fig. 1(b, e and h) also determine the local structure of the NCM811, which can be indexed to (110), (113) and (003) planes view along [1-10] direction, (102), (006) and (-104) planes along [010] direction, (104), (110) and (01) planes marked along [-441] direction.

2.2. Electrochemical performance of NCM811

Fig. 2(a) presents the NCM811 electrode delivered an initial discharge capacity of $200.4 \text{ mA h g}^{-1}$ with cut-off voltage of 4.3 V at 0.1C, and the capacity slightly increased to 210.1 and $219.2 \text{ mA h g}^{-1}$ upon increasing the cut-off voltage to 4.7 and 4.9 V, respectively. Fig. 2(b) is the galvanostatic charge/discharge profiles of NCM811-cycled at cut-off voltages of 4.3, 4.7 and 4.9 V, respectively. The NCM811 delivers a capacity retention of 95.40% after 50 cycles under 4.3 V operation, and the discharge capacity retention decreased to 80.15% and 69.05% under 4.7 and 4.9 V operation respectively, indicating the cycling stability of NCM811 is related to the cut-off voltage strikingly. Also, the NCM811 shows obvious capacity decay with the increase of charge cut-off voltage. As expected, charging to higher cut-off voltages can indeed maximize lithium utilization in the first several cycles, but the higher cut-off voltage also results in the deteriorated cathode due to the drastic reactions between cathode and electrolyte, which contributes to the rapid capacity fading of the cathode. EIS was carried out to investigate the interfacial electrochemistry and reaction kinetics of the NCM811-cycled at different cut-off voltages (Fig. S5 in Supporting Information). The OCV potentials of the samples for cycle test under cut-off voltages of 4.3, 4.7 and 4.9 V are 2.71, 3.12 and 3.22 V. Fig. S5(a and b) shows the EIS results of NCM811-cycled at different cut-off voltages after 50 cycles. It can be seen that these three impedance spectra are composed of two semicircles in high-frequency and intermediate-frequency. The impedance spectra were fitted using the equivalent circuit illustrated in Fig. S5(c), and the fitted results are listed in Table S3. In the equivalent circuit

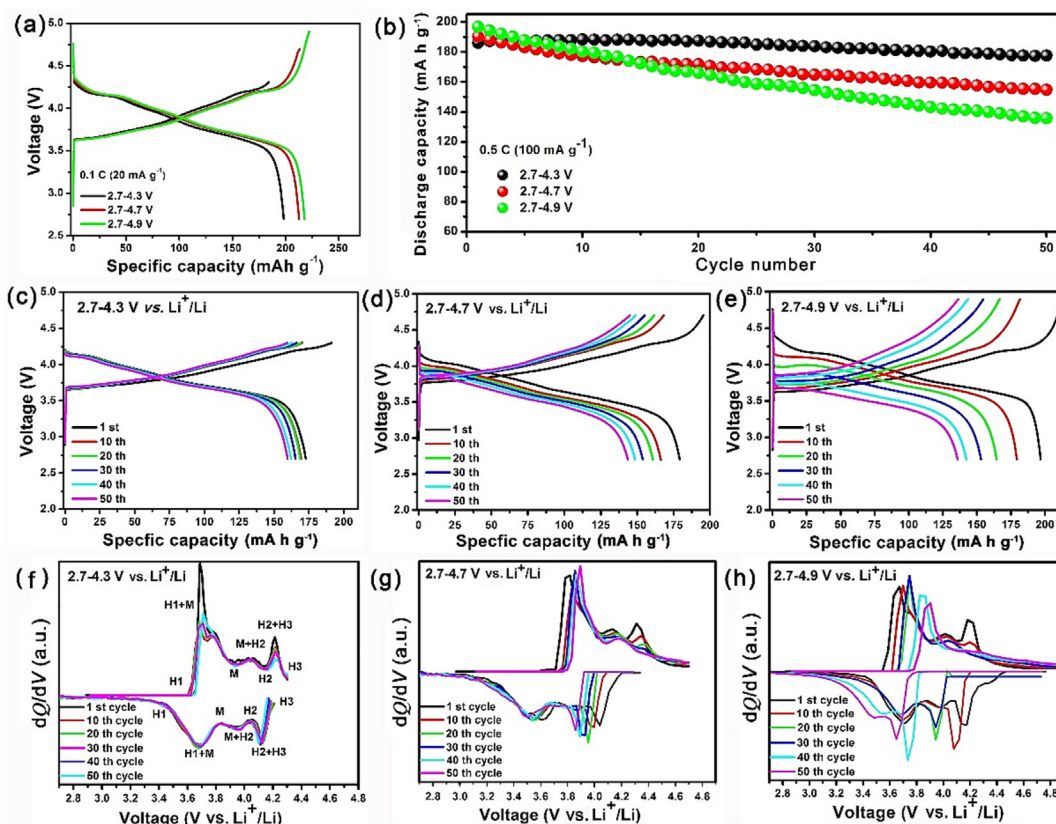


Fig. 2. Electrochemical performances of NCM811 under different cut-off voltages. (a) The initial charge–discharge, and (b) the cycling performance at cut-off voltages of 4.3, 4.7, and 4.9 V, respectively. The selected Galvanostatic charge/discharge profiles of NCM811 at cut-off voltages of (c) 4.3 V, (d) 4.7 V, and (e) 4.9 V. The dQ/dV plots of NCM811 at cut-off voltages of (f) 4.3 V, (g) 4.7 V, and (h) 4.9 V.

diagram, the two semicircles are attributed to R_{ct} and R_f . R_f is the solid electrolyte interface (SEI) layer impedance, which corresponds to the semicircle in the medium frequency region. R_{ct} represents the charge-transfer resistance of the cathode/electrolyte interface, and it is related to the second semicircle in the intermediate frequency region [24,25]. The sloping line in the low frequency region is associated with the Warburg impedance (Z_w) refers to the Li-ion diffusion in the electrode material. It is clear that R_{ct} value is 1292 Ω after 50 cycles at 4.3 V, 1319 Ω at 4.7 V and 1789 Ω at 4.9 V. The highest R_{ct} value at a cut-off voltage of 4.9 V may be ascribed to the most serious structural deformation that occurred for the electrode cycled at 4.9 V. The smaller R_{ct} of NCM811 cycled at 2.7–4.3 V after 50 cycles indicated the faster Li^+ ions transport (Table S3 in Supporting Information).

Fig. 2(c–e) is the selected galvanostatic charge/discharge profiles of NCM811 cycled at cut-off voltages of 4.3, 4.7 and 4.9 V, respectively. At 4.3 V, the NCM811 delivers negligible capacity fading and voltage decay, with the increase of cut-off voltage, more lithium ions are extracted, and the capacity gradually increases, but the voltage and capacity decay faster, as shown in Fig. 2(c–e). In addition, the polarization was intensified with the voltage increasing, which may be ascribed to the poor structural stability of NCM811 under the high cut-off voltage of 4.9 V resulting from the severe parasitic electrode–electrolyte reactions. Phase transition of the cathode material is one of the main reasons for cell degradation, which can be electrochemically explained by the dQ/dV analysis [26]. There are three distinct redox peaks that can be assigned to the complicated phase transitions evolving hexagonal (H1, H2 and H3) and monoclinic (M) phases [27]. The phase transition of H2-to-H3 phase is related to the large anisotropic lattice volume changes, resulting in cracks

and hindering Li re-intercalation into the system [28,29]. In Fig. 2(f), the NCM811 displays the negligible change of the peak intensities, no dQ/dV peak disappears from the first cycle to the 50th cycle, and the overlap of dQ/dV profiles within 50 cycles suggest the good reversibility of H2-H3 phase transition at cut-off voltage of 4.3 V. However, at 4.7 V, upon cyclings, significant capacity fading and voltage decay were observed [30–32]. As the cycle increases, the voltage polarization increases rapidly, the peak intensity decreases faster than the sample under cut-off voltage of 4.3 V due to the increased irreversible phase transition in the cathode material at a higher voltage (Fig. 2g). In addition, the decrease in peak intensities of H2-to-H3 phase transition indicates the fast capacity fading. These results suggest that higher voltage operation results in larger structural change and surface degradation, the cell cycled at 4.9 V exhibited more severe capacity fading and voltage decay upon electrochemical cycling, as illustrated in Fig. 2(e and h).

According to previous studies, the Li^+ diffusion coefficients (D_{Li^+}) is an important parameter of cathode materials for evaluating the rate of ionic or electronic transport within the cathode. The GITT experiments were carried out to determine the chemical diffusion coefficient of Li^+ [33], and to further investigate the change of Li^+ diffusion under different cut-off voltages. Fig. S6 in Supporting Information is a single step of GITT curve during the first charge process. Fig. S7 gives the GITT curves obtained during the first charge/discharge process of NCM811 cycled under different cut-off voltages. The D_{Li^+} can be calculated by the second Fick's law, the equation can be expressed as [34]

$$D_{\text{Li}^+} = \frac{4}{\pi\tau} \left(\frac{m_B V_m}{M_B S} \right)^2 \left(\frac{\Delta E_s}{\Delta E_\tau} \right)^2 \left(\tau \ll \frac{L^2}{D_{\text{Li}^+}} \right) \quad (1)$$

In this equation m_B is the mass of cathode material, V_m is the molar volume of the active compound, S is the effective surface area of the electrode, M_B is the molecular weight, τ means the pulse current time, L is the thickness of the electrode, ΔE_s is the difference between the E_s and E_0 during a single-step GITT progress. The D_{Li^+} values can be calculated by the above equation, and the results are illustrated in Fig. 3. Similarly, the D_{Li^+} values first increase and then decrease during the charge process. Particularly, the D_{Li^+} values of NCM811 at the beginning of the charge and at the end of the charge/discharge are much lower than at other states. It suggests that the diffusion of Li^+ is very slow, and then the insertion of Li^+ will be blocked under higher current density, which may lead to lower Coulombic efficiency, sluggish kinetics and a poor rate capability for cycled NCM811 at higher current density, as shown in Fig. S8. However, the D_{Li^+} values of NCM811 cycled at 4.3 V is higher than that of NCM811 cycled at 4.7 and 4.9 V. This may be ascribed to the growth of the structural reconstruction layer on the NCM811 surface under higher cut-off voltage. It's worth noting that the diffusion coefficient decreases significantly when cell is charged to 4.15 V at different cut-off voltages especially at 4.7 and 4.9 V, ascribing to the transition into the H3 crystal structure narrows the Li^+ diffusion channels by decreasing the interlayer spacing [19]. To support our interpretation, XRD analysis was performed after discharged to 2.7 V and the results are given in Fig. S9. All the samples cycled at different cut-off voltages retained the layered structure of pristine NCM811, except for a few weak peaks between 20–25°, which can be attributed to the superstructure reflections of Li_2MnO_3 (space group: C2/m) phase, originating from the super-lattice ordering of Li and Mn in the TM layer of NCM811 [35]. (Fig. S9a). Particularly, the (003) reflection associated with the c-axis interlayer spacing was changed visible [36]. Compared with the pristine NCM811, the position of (003) reflection of NCM811 cycled under 4.3 V shifts to a larger angle and gives a smaller interlayer spacing, which is due to the oxygen loss and the collapse of some layered structures [34]. After cycling at the cut-off voltage of 4.7 V, the position of (003) reflection shifts to a smaller angle and shows a larger interlayer spacing. This is due to the stronger repulsive force of the excessive delithiation layer [33]. After cycling at a higher cut-off voltage of 4.9 V, compared with 4.7 V, (003) peak shifts to a larger angle and displays a smaller interlayer spacing. This is because more excessive metals occupy the lithium sites, and the repulsion force between lithium layers was offset [37,38]. The samples cycled under 4.7 and 4.9 V are delithiated deeply, the interlayer spacing of NCM811 increases to a certain extent, even under larger cut-off voltage it can not be further expanded. Rietveld refinement results of the samples are tabulated in Table S4. Lattice constants of 'a' and 'c' stand for the distance between neighboring metal ions in the TM layer and the Li slab, respectively. Both of them displayed enlarged values for all the samples. The increase in the lattice parameter in

cycled samples under higher cut-off voltage provides direct evidence for the expansion of the Li layer, which is beneficial for the Li diffusion between oxygen layers. In addition, compared with the pristine NCM811, a decrease in the $I_{(003)}/I_{(104)}$ ratio for the cycled sample under higher cut-off voltage implies a transformation from a layered structure to a rock-salt phase.

The similar characteristics of the samples cycled under 4.7 and 4.9 V observed in XRD are related to the lithium-ion diffusion coefficient in GITT. As well known that the larger the interlayer spacing, the easier the lithium ion diffusion, and the greater the lithium ion diffusion coefficient, but in the GITT experiments, the diffusion coefficients of the samples cycled under 4.7 and 4.9 V are smaller than that of 4.3 V. This suggests that the increased interlayer spacing under higher cut-off voltage does not improve the lithium ion diffusion coefficient, which is reflected in the electrochemical performance of the battery, that is, the rate performance after cycling is still lower than 4.3 V. Hence, it can be concluded that after cycling under higher cut-off voltage, the cathode material particles undergo much severer polarization, strain and cracks. The larger changes of peak intensity and position in H2-H3, as shown in dQ/dV curves, also confirmed the results [35,39–41]. In addition, the higher cut-off voltage means that the reactions between the electrolyte and the cathode material are more violent [39], resulting in the formation of more spinel and rock-salt phase, which could block the diffusion of Li^+ [42,43].

2.3. Structural transformation

Pulverization and cracks are considered major contributors to the capacity decay of the Ni-rich cathodes [42]. To observe the pulverization of the cathode material charged to a higher cut-off voltage during cycles, the pristine electrode materials and the discharged samples after 50 cycles were performed by FE-SEM. The NCM811 microspheres maintain their pristine morphology without obvious pulverization though they were charged to a higher cut-off voltage of 4.9 V, as shown in Fig. S10. Usually, the Ni-rich NCM prepared by the co-precipitation method is a secondary particle formed by agglomeration of primary particles [44]. Because the primary particles have different crystal orientations, the crystal volume expands and contracts anisotropically during the charge and discharge processes. Microcracks are generated in the secondary particles, resulting in deterioration of electrochemical performance. To further illustrate the structure evolution of NCM811 under different cut-off voltages, cross-sectional specimens were conducted using FIB-SEM to check the detailed investigation inside STEM, as shown in Fig. S11.

According to previous studies, the main reason for the capacity fading of Ni-rich cathode is structural degradation which is specifically caused by an intergranular fracture due to the interface mechanical stress [45–47]. It can accelerate the side reactions

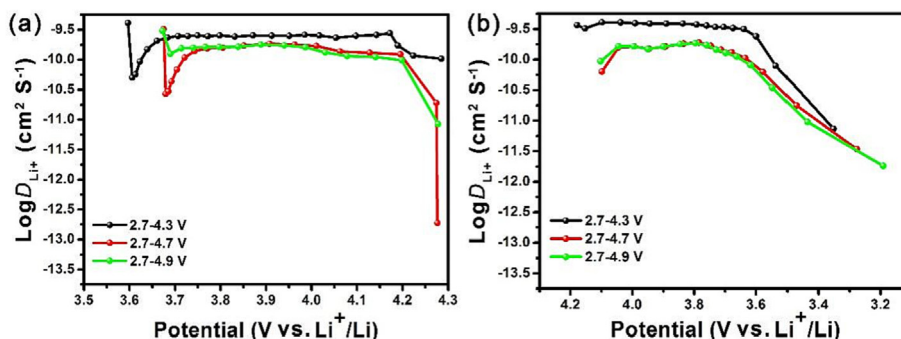


Fig. 3. The diffusion coefficients of Li^+ as a function of potential for NCM811 under 2.7–4.3 V, 2.7–4.7 V and 2.7–4.9 V after 50 cycles (a) charge, and (b) discharge.

between the electrolyte and the formed microcracks of cathode materials, and finally result in the failure of materials. Cracks can be divided into two types: intergranular crack and intragranular crack, both of which will damage the structure of cathode. The general features of these intergranular and intragranular cracks are identified in the cross-sectional STEM-HAADF images in Fig. S11. It is obvious that after 50 cycles, the samples cycled at cut-off voltages of 4.3 and 4.7 V exhibit no significant differences in terms of the intergranular cracking features (Fig. S11b and c), only the lattice distortion stage can be observed on the particle surface at the cut-off voltage of 4.3 V, as shown in Fig. S12 [48]. However, more intragranular and intergranular cracks appeared in the NCM811 cycle with a cut-off voltage of 4.9 V, as shown in Fig. S11(d). The intragranular cracking characteristics perhaps substantially contribute to the faster capacity degradation cycled at 4.9 V as compared with those cycled at 4.3 and 4.7 V. Hence, we focused our effort on the sample cycled at different cut-off voltages to reveal the detailed structure of the intragranular cracks and understand their formation mechanism.

To reveal the structural evolution and the microstructural degradation of the NCM811 under different cut-off voltages in detail, extensive STEM-HAADF and ABF analyses were performed at the atomic scale. All these STEM images were filtered using an Average Background Subtraction Filter (ABSF) method. Fig. 4 presents the phase transformation of NCM811 cycled at 4.3 V. The pristine NCM811 has a layered structure both on the surface and in the bulk primary particle. The layered phase has a hexagonal structure with a space group of $R\bar{3}m$ ($a = b = 2.877 \text{ \AA}$, $c = 14.228 \text{ \AA}$), where the TM and lithium occupy octahedral 3b and 3a sites, respectively [15]. The layered structure was further confirmed using atomic-resolution HAADF-STEM and ABF images, as given in Fig. 4(d and e). In addition, the line profiles of TM layer and lithium layer are labeled with the red and green arrows in HAADF images and the corresponding structure models are shown. The NCM811 evolves the phase transformation from the layered feature (Fig. 4a) in the bulk to rock salt phase (Fig. 4c) in the surface, accompanied by the intermediate phases of disordered layered phase and defect rock salt phase (Fig. 4b and c). A series of HAADF and ABF images were captured to give more details of the phase transformation. Fig. 4(d and e) are the HAADF and ABF images of the layered structure of NCM811 along $[10\bar{1}0]$ zone axis, the TM atomic columns can be confirmed from the $R\bar{3}m$ structure model simulated in Fig. 4(g), showing that no TM ions detect in the lithium layers and thus smooth lithium ion diffusion channels. The line profiles (Fig. 4f) of TM and lithium layers also indicate the lithium ions sites have not been occupied by TM cations. As demonstrated in Fig. 4(c), some atomic columns appeared between two TM layers displaying an obvious interface between the two phases (marked by the red line), indicating that during repeated charge/discharge cyclings some TM cations migrated into lithium ion diffusion channels, which are determined by the HAADF and ABF images in Fig. 4(h and i). Also, the line profiles in Fig. 4(j) support the existence of TM cations. Obviously, the migration of TM cations leads to the formation of a disordered layered structure NCM811 (Fig. 4k). Then serious migration of TM cations gives a defect rock-salt layer, as demonstrated in Fig. 4(l–o). The line profiles suggest some Li sites still can be observed. Finally, the $Fm\bar{3}m$ rock-salt NiO phase is clearly observed, saying the lithium ion transportation pathways are fully occupied by TM cations (Fig. 4p–s). Such structural degradation of NCM811 during cycles maybe result in the capacity fading, sluggish kinetics and poor rate performance. The above results indicate that cation mixing will occur on the particle surface after a short cycle under the cut-off voltage of 4.3 V. Therefore, it is very important to find a way to avoid cation mixing to improve the electrochemical performance of NCM811.

Furthermore, the microstructure evolution of NCM811 after 50 cycles under a higher cut-off voltage of 4.7 V was also investigated by the atomic resolution Z-contrast HAADF-STEM and ABF-STEM images. As shown in Fig. 5, more abundant phase transitions are observed. In the bulk, well layered structure was maintained (Fig. 5b), with all TM ions located at TM layers. Lithium is unable to be seen in the atomic-scale HAADF image in state-of-the-art. However, in the Fig. 5(c and d), the contrast in Li layer appears, it proves that partial TM cations have gradually migrated from TM octahedral sites into Li octahedral sites, as signed by the dashed circle (Ni_{Li}), resulting in the formation of disordered layer (Fig. 5c and d). The spinel (Fig. 5e) and defect rock salt (Fig. 5g) were observed closer to the surface of the particles, and more TM cations migrated into the Li octahedral sites. On the surface of the particles, the TM ions completely blocked the diffusion channels of lithium ions, and the rock-salt phase (Fig. 5h) appears.

The structural change of NCM811 cycled at cut-off voltage of 4.9 V was examined using atomic-level HAADF-STEM. As shown in Fig. 6(a), intergranular cracks and intragranular cracks are obviously observed in the sample cycled at 4.9 V (marked by red and yellow arrows), but not seen in the samples cycled at 4.3 and 4.7 V. Therefore, it is speculated that the intragranular cracks are the main contributor to the capacity fading under higher cut-off voltage. During lithium ion extraction and insertion, the crystal lattice will expand and shrink. In addition, out of our expectation, disorder layered, spinel, and defect rock salt was not observed, only layered and rock salt structures (Fig. 6b–d) are observed, which are different from the results captured on the samples cycled at 4.3 and 4.7 V. The reason is that charging too high voltages of 4.9 V means maximizing lithium utilization in the first several cycles resulting in deep delithiation, and more Ni cation migration from the octahedral TM layer to the Li layer, which leads to the rapid structural changes from layered to disordered spinel or rock-salt phase driven by the rapid reduction of Ni^{4+} to Ni^{2+} , concurrently accompanied by oxygen release. However, the spinel phase with a high Ni content is very unstable due to the mixture of unstable Ni^{3+} and Ni^{4+} , such that phase transition to rock-salt NiO is likely to occur for NCM811 under 4.9 V. After a careful scan of the sample cycled under 4.9 V, many holes, dislocations and cracks are appeared (Fig. 6e–g). The formation of holes may be ascribed to the easier oxygen release upon deep delithiation (Fig. 6e), implying the instability of NCM811 under higher cut-off voltage. Fig. 6(f) shows dislocations in atomic resolution, near which intragranular are discovered clearly, suggesting the dislocations are the origin of intragranulars. For intragranular cracks, the two free surfaces appear to be parallel along the whole crack, which is markedly different from an edge-shaped crack formed by fast extension of crack under mechanical stress. Furthermore, all intragranular cracks follow a strict crystallographic preference along the (003) layered plane (Fig. 6g and h), there are no atoms in the crack, as can be confirmed from the profile curve in Fig. 6(h), which still possesses internal layered structural features, and the width of the premature crack is about 2.7 nm. We noted that the intragranular cracks will finally evolve into premature cracks, as illustrated in Fig. 6(g) and Fig. S13 in Supporting Information. As well known, the premature crack may further develop into a real intergranular crack during the repeated cycling of the battery, and lead to a negatively impact on the cycling performance [27,30]. The intergranular cracks were regarded as a major contributor to the electrochemical performance fading during cyclings. Interestingly, phase transformation is also recognized near intergranular, as confirmed in Fig. 6(i). In the atomic-level HAADF-STEM images, layered structure (Fig. 6j and k) and rock-salt phase (Fig. 6m–o) is verified. The lattice spacing of 4.7 Å in Fig. 6(k) could be ascribed to the (003) plane in the layered structure of NCM811 phase.

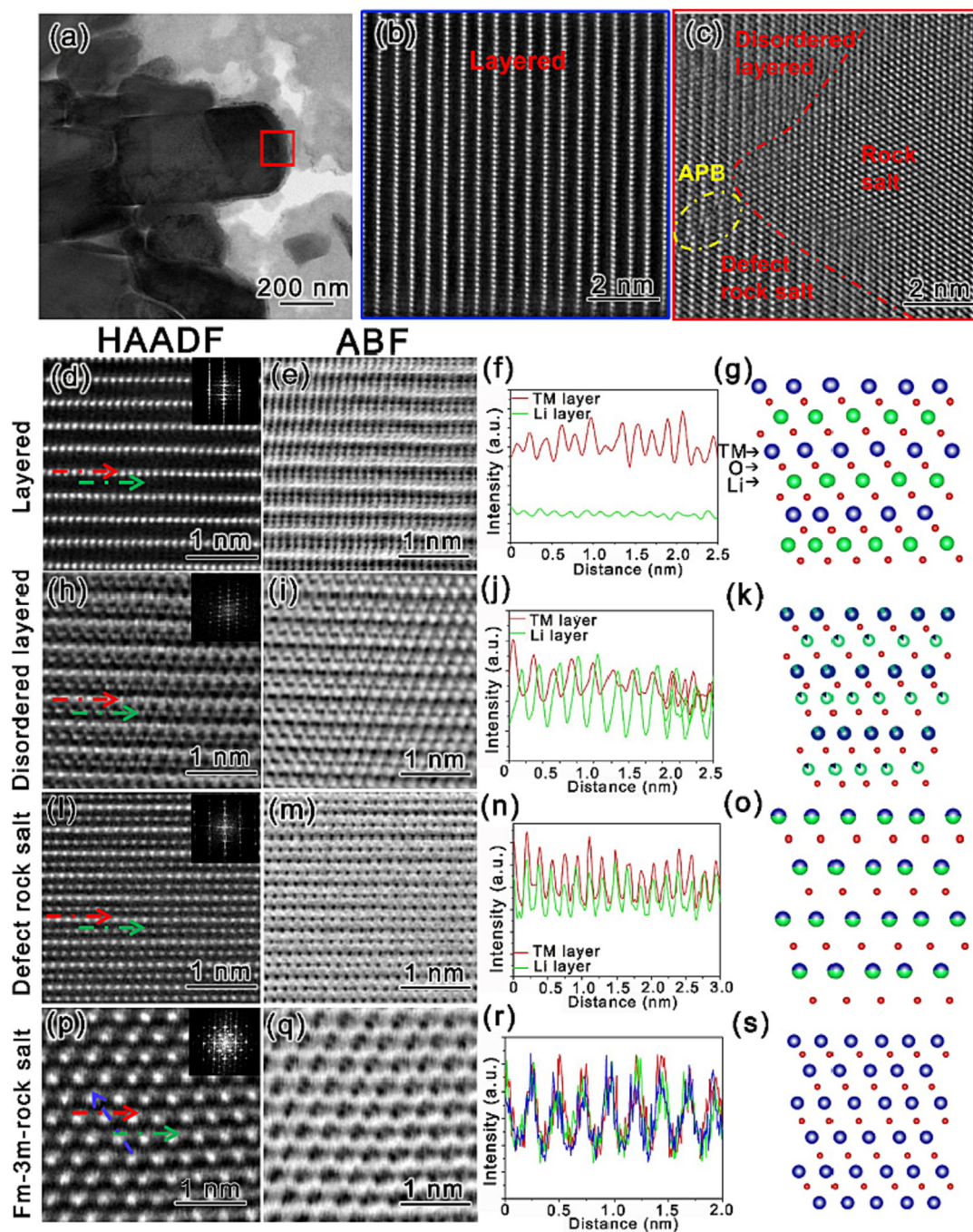


Fig. 4. Atomic-resolution STEM-HAADF and ABF images captured on NMC811 cathode after 50 cycles at a cut-off voltage of 4.3 V in [110] zone projection. (a) Cross sectional low magnification ABF image, (b and c) STEM-HAADF images at different regions from bulk to surface. The red dashed lines denote the surface reconstruction layer at different stages. Detailed structural information from (d and e) layered region, (h and i) disordered layered region, (l and m) defect rock salt region, and (p and q) rock salt region. (f, j, n, and r) Corresponding line intensity profiles of the layers labeled with the same color of arrows in HAADF images. (g, k, o, and s) Corresponding structure models for explaining the structure transition process.

Fig. 6(l and p) illustrates the structure models of layered phase (R-3 m) along [100] zone axis and rock-salt (Fm-3 m) viewed along [111] orientation.

Under the higher cut-off voltage of 4.9 V, almost all lithium ions participate in the extraction and insertion processes. Due to the H2-H3 phase transition, the anisotropic expansion and contraction of the primary particles are easy to occur [29,25], and then a large number of intergranular cracks are formed, which can destroy the contact between the primary particles, reducing the internal conductivity of the secondary particles, and finally produce an uneven

state of charge [46]. The electrolyte will penetrate into the intergranular cracks, which could result in side reaction, surface degradation, electrolyte consumption, and irreversible loss of Li^+ or electrolyte, and reduce the capacity of the material [12]. Due to the charge compensation effect, Ni^{2+} near the Li vacancy will be oxidized to Ni^{4+} [49]. Unfortunately, Ni and O also have a charge compensation effect, a large number of oxygen vacancies will be created, oxygen vacancies will lower the diffusion barrier of Ni^{2+} , and Ni^{2+} will more easily diffuse into the Li layer, which will lead to the formation of rock-salt phase, which blocks the lithium

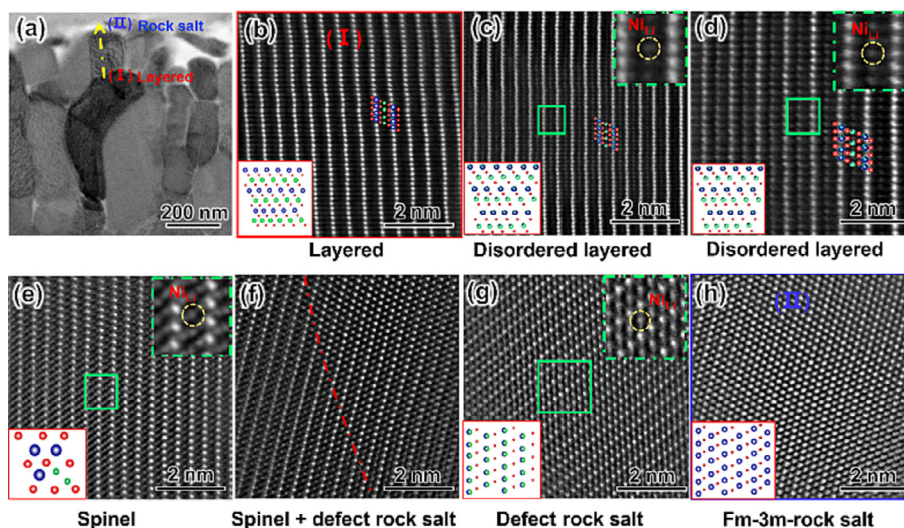


Fig. 5. Crystal structure evolution of NCM811 cathode after 50 cycles with cut-off voltage of 4.7 V. (a) The cross sectional low magnification ABF-STEM image and (b–h) the atomic-level STEM-HAADF images captured on NCM811 at different regions from bulk to surface. The gradually increasing Ni_{Li} in Li site is marked by a dashed circle.

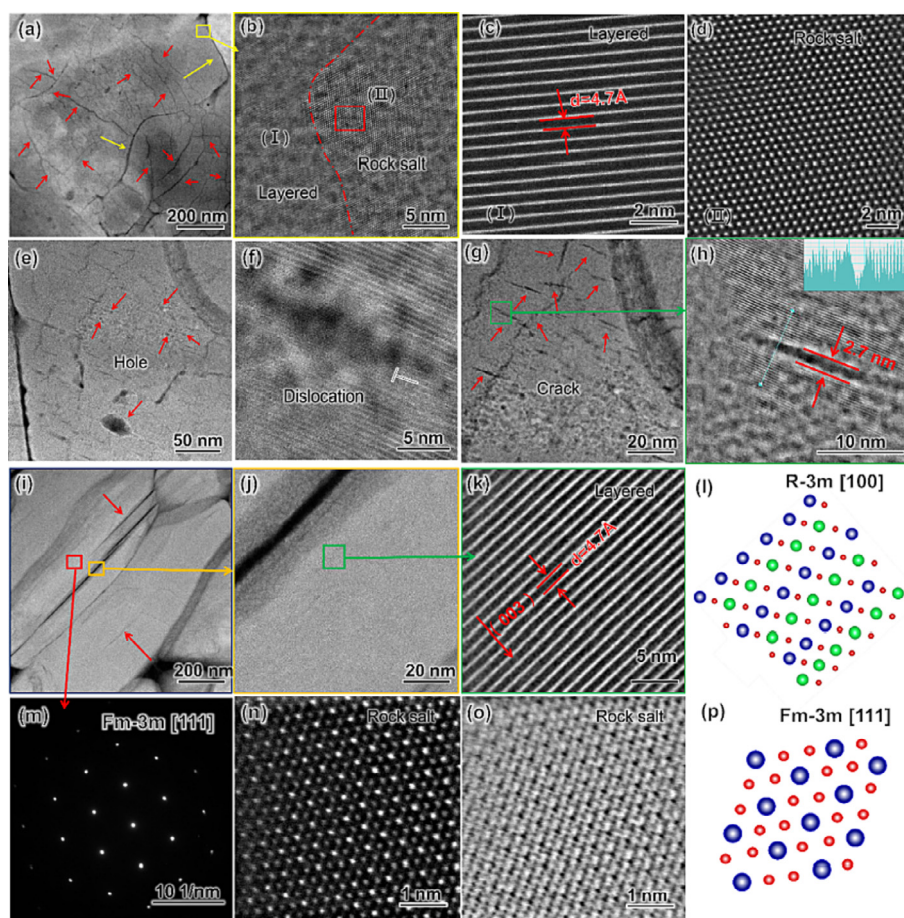


Fig. 6. Crystal structure evolution of NCM811 cathode after 50 cycles with cut-off voltage of 4.9 V. (a and e) Cross sectional low magnification HAADF-STEM image of NCM811 after 50 cycles at cut-off voltage of 4.9 V. (b) The corresponding HAADF-STEM image of the marked yellow rectangle regions in (a). (c and d) The corresponding enlarged HAADF-STEM images of (b) marked as I and II. (f and g) The HAADF-STEM images indicate the dislocation and cracks in the sample. (h) The enlarged image corresponding to the green rectangle in (g). (i) The HAADF-STEM image shows the cracks. (j) The corresponding HAADF-STEM images of the yellow rectangle in (i). (k) The enlarged image corresponding to the green rectangle in (j). (l–o) The corresponding SAED pattern, HAADF-STEM and ABF-STEM images of red rectangle in (i). (l and p) The corresponding structure models of R-3 m and Fm-3 m.

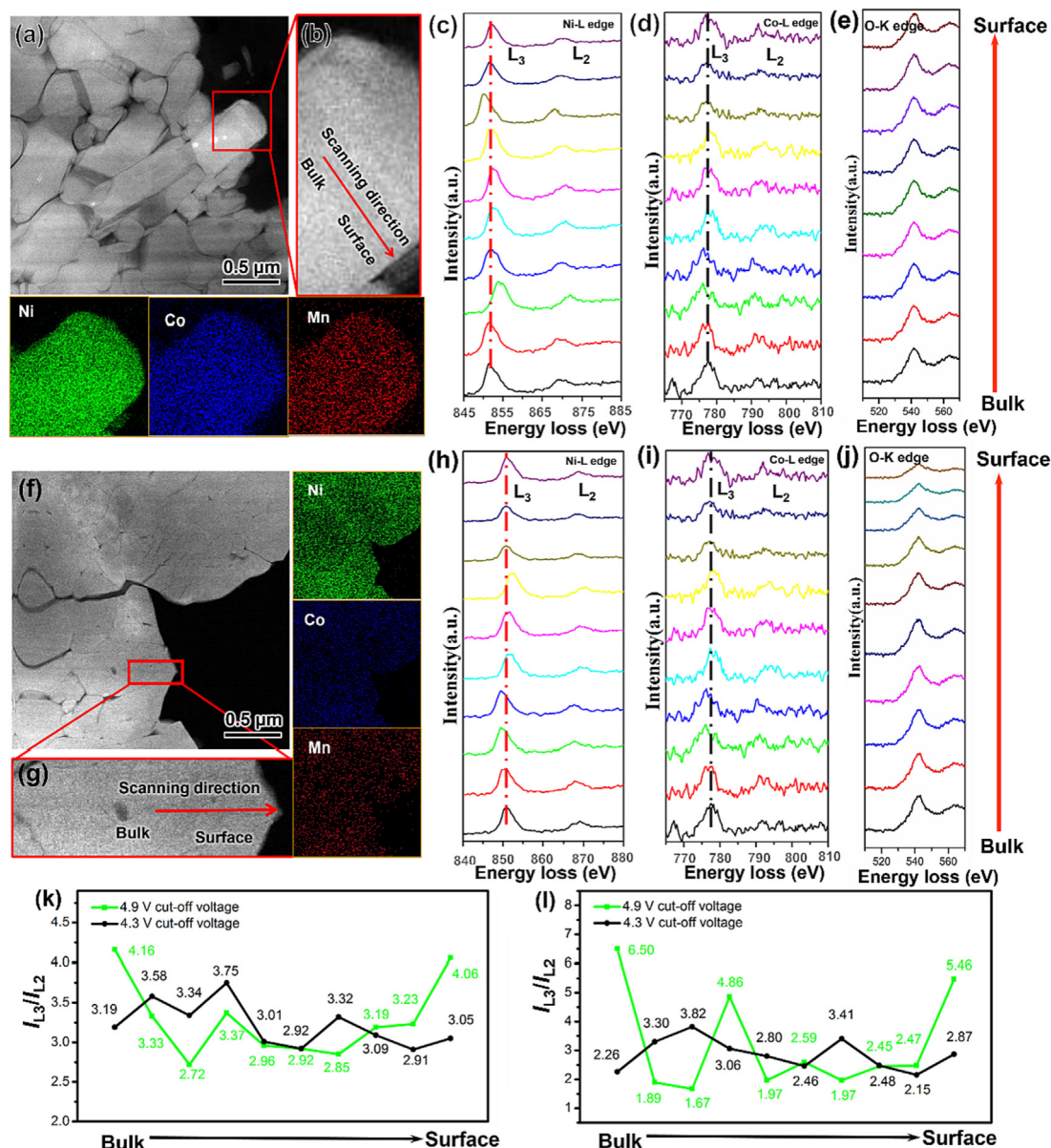


Fig. 7. The STEM-EELS spectrum images of NCM811 particles cycled under different cut-off voltages. (a and f) Cross sectional low magnification STEM-HAADF images and the elemental mappings of NCM811 after 50 cycles at cut-off voltages of 4.3 and 4.9 V. (b and g) The corresponding images of the marked red rectangle regions in (a and f), and the red arrows show the areas from where EELS line profiles have been collected. (c–e, h–j) The corresponding EELS line profiles of Ni-L, Co-L and O-K edges obtained in the directions in (b and g). (k and l) The estimated I_{L3}/I_{L2} of Ni (k) and Co (l) in the cycled NCM811 under 4.3 and 4.9 V as a function of distance from the bulk to the surface.

diffusion pathways and leads to a loss of electronic/ionic transport of Li ions at the electrode/electrolyte interface. As a result, cell internal resistance increases greatly and causes large voltage polarization, and some extent of capacity fading. At the same time, the rock-salt phase will cause an uneven state of charge, and thus cause the imbalance of material contraction/expansion, finally, the cracks are formed and destroy the structural stability of materials. In addition, during Li^+ extraction and insertion, the adjacent particles of different phases will produce stress concentration at the phase interface, and the force will be generated at the grain boundary of the two primary particles, resulting in cracks along the grain boundary. The particle cracking and pulverization results in a loss of electronic contact within the composite electrode, causing the loss of cathode active material and increased cell internal resistance. Therefore, it is very important to suppress intergranular cracks under high voltage. For NCM811, interface engineering modification is necessary when operating at a higher voltage.

Due to the limited field of view with the atomic-scale STEM-HAADF image, it is difficult to analyze the changes across the entire primary particles. The STEM-electron energy-loss spectroscopy (EELS) spectrum images were collected from bulk to the surface of NCM811 particles to understand the phase transformation of the cycled cathode (50 cycles at 0.5C) at the discharge state. According to previous reports, TM L-edge was an indicator for the oxidation states of TM ions in the layered cathodes [50], EELS edges of TMs exhibit L_3 and L_2 white lines, which result from the transition of electrons from the spin-orbit split levels $2p_{3/2}$ and $2p_{1/2}$ to unoccupied $3d$ states. The relative intensity of L_3 and L_2 white lines (I_{L3}/I_{L2}) and energy position are related to the d-band occupancy and valence state of the TMs. Fig. 7(a and f) are the STEM-HAADF images and the corresponding elemental mappings of NCM811 after 50 cycles under cut-off voltages of 4.3 and 4.9 V. Line profiles were acquired in the direction of the red arrows shown in Fig. 7(b and g). Ten spectra were collected from the bulk

to the surface, each one averaging over 10 nm, and the interval was 40 nm. As shown in Fig. 7(c and d) (4.3 V cut-off voltage) and Fig. 7(h and i) (4.9 V cut-off voltage), the Ni and Co-L edges spectrums both consist of two major peaks, corresponding to the $2p_{1/2}$ orbitals (L_2 edge) and $2p_{3/2}$ orbitals (L_3 edge). The change of I_{L3}/I_{L2} in the Ni and Co valence state under different cut-off voltages of 4.3 and 4.9 V are summarized in Fig. 7(k and l). The I_{L3}/I_{L2} of Ni becomes higher at the surface (Fig. 7k), which implies a decrease of valence at the surface due to the reduction of transition metals Ni. In other words, the generation of the Ni^{2+} phase is accelerated and propagated from surface to the bulk. In addition, higher I_{L3}/I_{L2} of Ni under 4.9 V are observable both in the bulk and surface regions. This region coincides with cracked areas inside the bulk. These results also indicate that more cracked areas are formed under higher cut-off voltage. The migration of transition metal ions to the lithium layer, and the reaction between electrolyte and Ni^{4+} lead to the formation of rock-salt phase driven by the rapid reduction of Ni^{4+} to Ni^{2+} . The formation of rock-salt phase blocks the lithium diffusion pathways, causes capacity fading and reduces the lithium ion diffusion coefficient, and thus has a negative impact on the cycle and rate performance of NCM811.

It shows that there is no O pre-peak in the spectra obtained in bulk and surface of both two samples. The disappearance of the pre-peak implies the loss of oxygen and the formation of oxygen vacancies in the materials [51]. Saying that the loss of oxygen and formation of oxygen vacancies are both observed on the surface of material under cut-off voltages of 4.3 or 4.9 V. It is worth noting that the peak intensity of oxygen gradually decreases on the material surface after 4.9 V cycling, indicating that more oxygen loss occurred at a higher cut-off voltage of 4.9 V. In addition, the large amount of released oxygen would create a significant number of oxygen vacancies [52], thereby lowering the activation barrier for the migration of TM cations, and eventually accelerating the phase transitions, which produces a large strain and severe microcracks. The above results are well consistent with the STEM results. Hence, we can draw a conclusion that more cracks are formed due to the more critical reaction between cathode material and electrolyte at a higher cut-off voltage. For Co L-edges, there is no obvious chemical shift can be observed, as shown in Fig. 7(d and i). However, the I_{L3}/I_{L2} of Co under 4.9 V has changed severely, which implies Co has involved in the redox reaction and contributes capacity under higher cut-off voltage.

3. Conclusions

To summarize, the structural evolution mechanism of NCM811 at atomic scale under various cut-off voltages was proposed by resorting to STEM-HAADF imagings and EELS spectra. Our results demonstrated that the crystal structure of NCM811 evolution from a layered structure to a rock salt phase was directly accompanied by serious intergranular cracks under 4.9 V, which is distinguished from the structure evolution of layered structure, disordered layered, defective rock-salt and rock-salt phases under 4.3 and 4.7 V. When charged to a higher cut-off voltage of 4.9 V, more defects, such as holes, dislocation and intragranular cracks, are observed, which cause intergranular cracks during repeated cycles. The newly formed intergranular cracks will consume more electrolytes. As a result, the newly exposed surface resulting from the intergranular cracks gradually propagates from the surface to the bulk, also the reduction of Ni and Co evolves the structural degradation of NCM811, which finally produce the capacity fading. Our work gives the degradation mechanism of NCM811 under high cut-off voltage, and provides insights for achieving high energy LIBs using NCM layered cathodes.

Experimental section

Experimental details can be found in the [Supporting Information](#).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jechem.2022.11.016>.

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